

Exam Halls & Proctoring Allocation Management System

Graduation Project - Technical Documentation - v1.0.0
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1. Executive summary

1.1 Problem

Before every exam session a school's exam-control office must, by hand:

- seat hundreds of students in rooms whose exam capacity (with spacing) is lower than their normal capacity;
- assign a head of committee and enough proctors to every room;
- respect conflict-of-interest rules — a teacher may not supervise an exam of their own subject, nor any session in which a relative (up to the 4th degree) is sitting;
- spread the supervision load fairly and respect each teacher's daily and weekly limits;
- react within minutes when a proctor is absent on exam day, and keep a trace of every change;
- print desk stickers, call sheets, schedules and a compensation report.

Doing this on paper is slow and error-prone, and rule violations are only discovered on exam day.

1.2 Solution

A desktop application for the exam-control office in which **the rules live in the database**:

Capability	How
Automated seating	<code>pkg_seating.generate_seating</code> — all students of the exam's grade, capacity-aware, special-needs students in front, room sharing between simultaneous exams
Fair proctor allocation	<code>pkg_supervision.allocate_proctors</code> — 1 head + 1 proctor per 20 students per room; least-loaded teacher first
Conflict-of-interest prevention	Subject conflict (per room), time conflict, daily / weekly caps, relatives (per session, any room) — checked for every assignment
Emergency 1-click substitution	<code>pkg_supervision.replace_proctor</code> — best eligible teacher, hours transferred, immutable audit record
Safety interlocks	Triggers refuse archiving teachers/rooms still in use, shrinking a room that has future seating, editing the audit log
Reports	Seating stickers, attendance call sheets, individual schedules, daily control sheet (PDF) and hours-for-compensation workbook (Excel)
Bilingual UI	Every screen, message and report in English and Arabic, right-to-left layout in Arabic

1.3 Scope delivered

Spec screen	Screen in the application	Roles
1	Login (+ Users & Roles management)	everyone / School Admin
2	Teachers & Staff (with relatives register)	Admin, Control Officer
3	Rooms & Halls	Admin, Control Officer
4	Seating & Supervision — seating plan	Admin, Control Officer (Committee Head read-only)
5	Exam Schedule (with exam creator/editor)	Admin, Control Officer (Committee Head read-only)
6	Seating & Supervision — supervision matrix	Admin, Control Officer (Committee Head read-only)
7	Emergency Substitution dialog	Admin, Control Officer
8	Occupancy Grid	Admin, Control Officer, Committee Head
9	Reports & Exports	Admin, Control Officer, Committee Head
—	Executive Dashboard (analytics & operations center), Courses & Departments, Students, My Duties	see §5

1.4 Key figures

Item	Count
Database tables / views / triggers / PL/SQL packages	13 / 5 / 16 / 4
SQL + PL/SQL (scripts 00–05)	~1,800 lines
Java source files / lines	128 / ~9,800
FXML views	15
Translated messages (EN = AR)	468 each
Automated tests	84 JUnit test cases (64 unit, 20 integration) + 25 SQL smoke checks

2. System architecture

2.1 Technology stack

Layer	Technology	Version	Why
Language / runtime	Java	21 LTS (runs on 21+)	LTS on lab machines; records, switch expressions, text blocks
UI	JavaFX + FXML + CSS	21.0.6	Native desktop UI, declarative views, full RTL support
Database	Oracle Database	tested on Free 23.26 (26ai); scripts use 12c+ features	Required platform; PL/SQL keeps rules next to the data
JDBC driver	ojdbc11	23.6	Oracle-certified for Java 11+
Connection pool	HikariCP	6.2.1	Fast, small, production standard
Password hashing	at.favre.lib bcrypt	0.10.2	Adaptive hash, \$2a\$ compatible
PDF	OpenPDF	2.0.3	LGPL/MPL fork of iText 2 with Arabic shaping and bidi
Excel	Apache POI (poi-ooxml)	5.4.0	.xlsx with formulas and RTL sheets
Logging	SLF4J + Logback	2.0 / 1.5	Console + rolling file logs/examhalls.log
Build	Maven (+ shade, assembly, javafx plugins)	3.9	One command to run, test and package
Tests	JUnit 5	5.11	Unit + database integration tests

2.2 Layered architecture

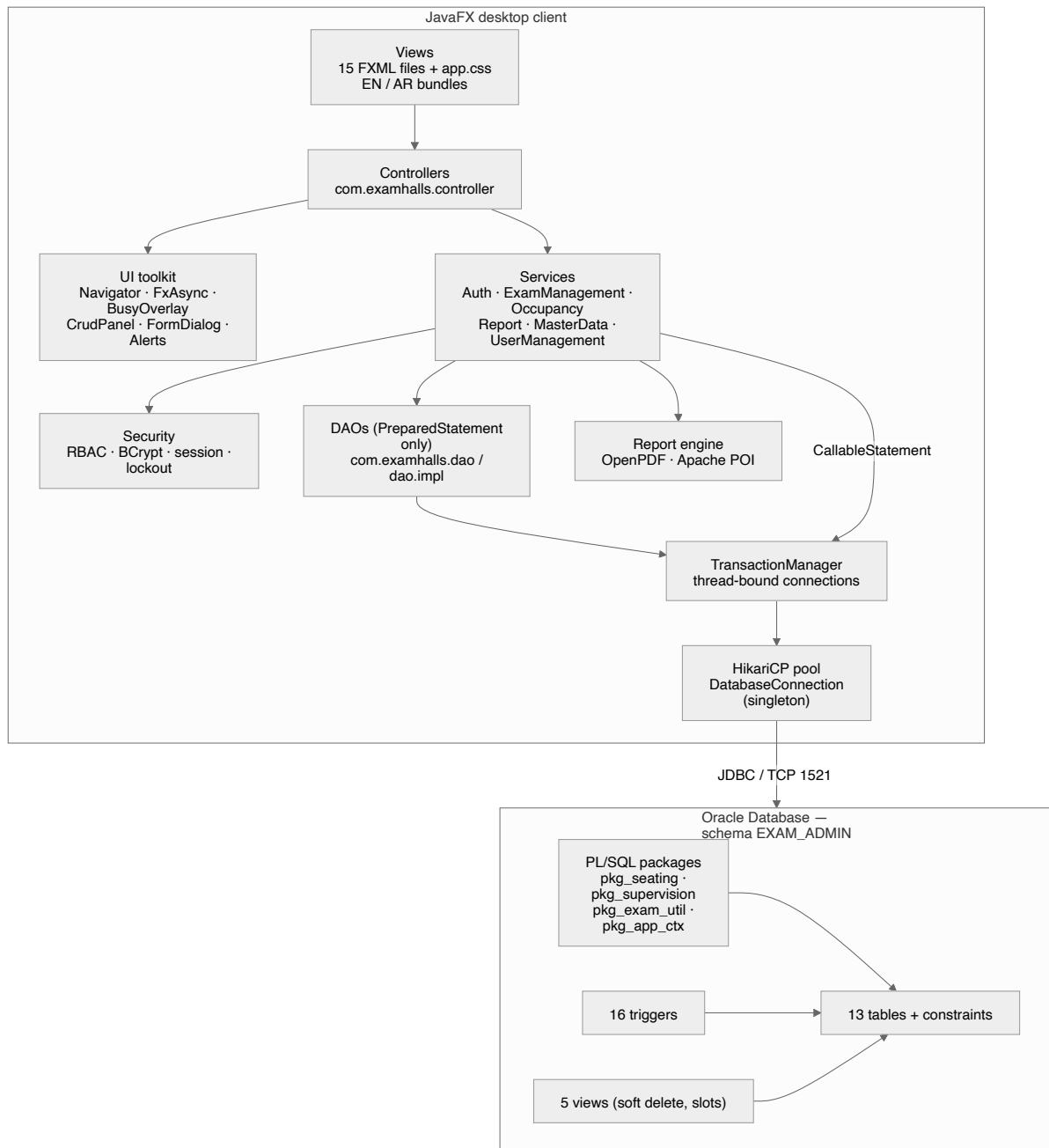


Figure 1 - Layered architecture

Pattern: MVC with a service layer and DAOs.

- **Model** — immutable Java records (Teacher, Room, ExamOverview, TeacherCandidate, ...) that mirror tables, views and procedure results; enums for every status column.
- **View** — FXML files for page structure, one shared stylesheet (app.css) with design tokens, and two resource bundles for all text.

- **Controller** — one per view; it binds records to controls and calls services on background threads. Controllers never touch JDBC.
- **Service layer** — enforces permissions, validates input, owns transactions and calls PL/SQL.
- **DAO layer** — plain SQL with `PreparedStatement`; no ORM, so the SQL shown in this document is exactly what runs.

MVVM was considered; JavaFX property bindings are used where they help (observable/filtered lists, selection listeners), but full view-models would duplicate the records without adding value for a project of this size.

2.3 Package structure

```
com.examhalls
├── MainApp, Launcher  entry points (Launcher: also --check-db)
├── config             DatabaseConnection (HikariCP), TransactionManager,
                       AppSettings
├── security           RoleType, Permission, UserSession, PasswordHasher,
                       LoginAttemptTracker
├── model              records and enums
├── dao                interfaces
├──   └── impl          JDBC implementations (JdbcSupport base class)
├── service            AuthService, ExamManagementService,
                       OccupancyService, ReportService, MasterDataService,
                       UserManagementService, DashboardService,
                       ServiceRegistry
├── report             PdfKit, PdfReports, HoursWorkbook, ReportFonts
├── exception          AppException, ErrorCode, OracleErrorTranslator
├── ui                 Navigator, View, FxAsync, BusyOverlay, Alerts,
                       Exports, CrudPanel, FormDialog
├── controller         one controller per screen / dialog
├── util               Messages (i18n), Formats (locale-aware formatting)

resources/ application.properties, logback.xml, view/*.fxml,
            css/app.css, i18n/messages[_ar].properties
```

2.4 Connection pooling (HikariCP)

`DatabaseConnection` is a **thread-safe, lazily created singleton** (volatile field + double-checked locking). Unlike the “holder class” idiom, a failed start (database down) can be retried on the next call.

Setting	Value	Reason
<code>maximumPoolSize</code> / <code>minimumIdle</code>	10 / 2	Desktop client: few concurrent statements
<code>connectionTimeout</code>	5 s	Snappy failure in the UI
<code>oracle.net.CONNECT_TIMEOUT</code>	4 s	Driver gives up before the pool, so the real ORA- error is reported
<code>initializationFailTimeout</code>	1 ms (one attempt)	Fail fast; never retry a wrong password in the background — Oracle's default profile locks an account after 10 failed logins
<code>maxLifetime</code> / <code>idleTimeout</code>	30 min / 10 min	Below typical firewall idle cut-offs
Statement cache / row prefetch	50 / 50	Oracle driver tuning
<code>v\$session.program</code>	<code>ExamHallsApp</code>	DBAs can identify the sessions

Externalised configuration, later sources override earlier ones:

1. `application.properties` inside the jar (defaults, **no password**)
2. `config/application.properties` next to the application, or `-Dapp.config=<file>`
3. environment variables `DB_URL`, `DB_USERNAME`, `DB_PASSWORD`
4. JVM system properties, e.g. `-Ddb.password=...`

Transactions. `TransactionManager` binds a connection to the current thread. DAOs call `tx.connection()`: inside `inTransaction(...)` they share the transaction's connection; outside they borrow an auto-commit connection. `runAndRollback(...)` executes real work and discards it — used by all integration tests.

2.5 Threading model

All database work runs off the JavaFX Application Thread through `FxAsync.run(work, onSuccess, onError, always)`. A `BusyOverlay` (a reference-counted spinner) covers the view and blocks input while a procedure runs, so the window never freezes and nobody double-clicks "Allocate".

2.6 Error-handling pipeline

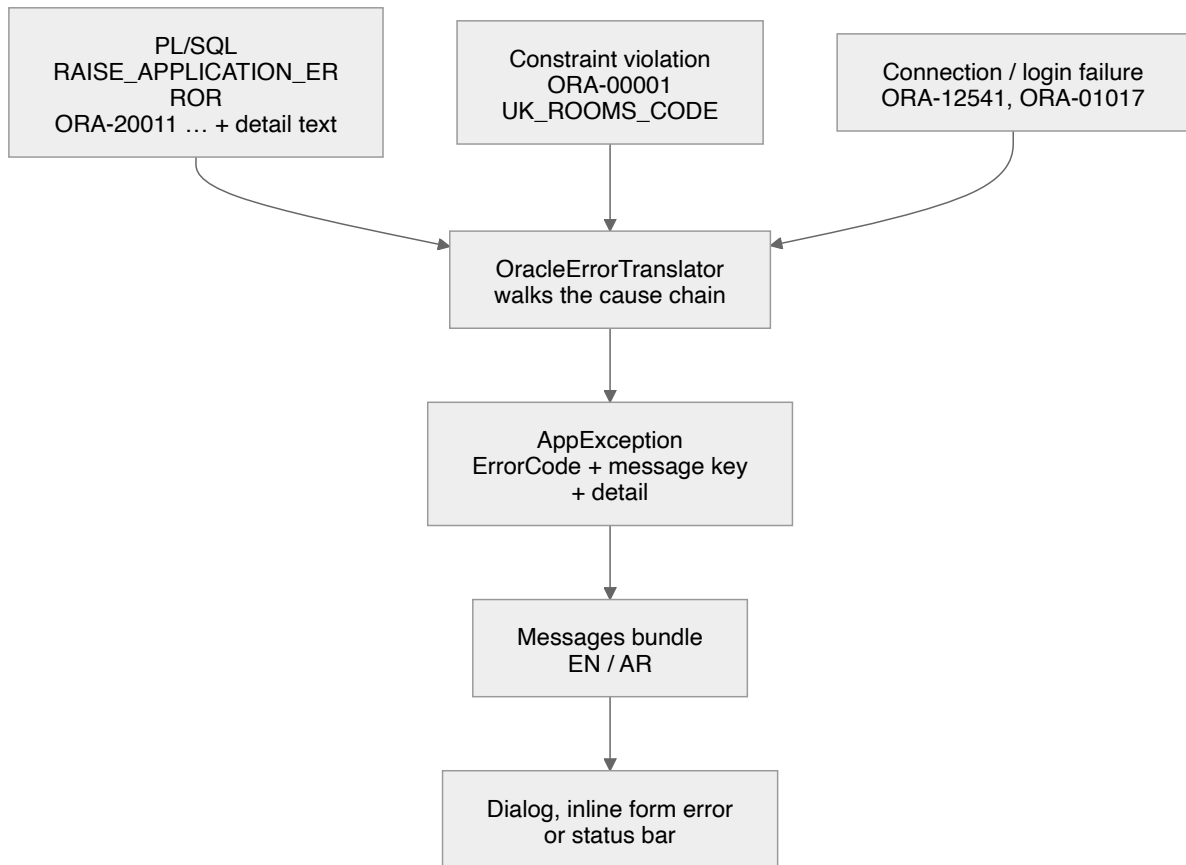


Figure 2 - Error-handling pipeline

- **Application errors** (-20001 ... -20030) map to `ErrorCode`s, and the PL/SQL text with its numbers (e.g. “300 students, only 248 free seats”) is kept as secondary detail.
- **Constraint errors** carry the constraint name, so the user sees a specific sentence such as “A room with this code already exists”. Foreign keys have two keys, `constraint.FK_x` (parent missing) and `constraint.child.FK_x` (still referenced).
- **Login errors** (ORA-01017, 28000, 28001) are reported as “account rejected”, not “network down”.
- A unit test ensures every key exists in both English and Arabic.

2.7 Security

Measure	Implementation
Password storage	BCrypt cost 12 (<code>\$2a\$</code>), CHECK constraint on <code>USERS.PASSWORD_HASH</code> format
No user enumeration	Unknown username and wrong password give the same message and take the same time (a dummy BCrypt verification runs for unknown users)
Input validation	One <code>CredentialPolicy</code> for client and server. Usernames must match <code>^[A-Za-z0-9._-]{3,50}\$</code> (no spaces, quotes, SQL or control characters). The login screen caps the fields while typing (50 / 128 characters) and rejects a malformed username at once; <code>AuthService</code> repeats every check before any database access . Malformed input costs the same time as a wrong password and counts as a failed attempt
Whitespace	Leading/trailing whitespace around the username is removed with <code>strip()</code> (Unicode whitespace only, so a hidden control character is rejected rather than silently removed). Passwords are never trimmed: spaces are valid password characters
BCrypt 72-byte limit	BCrypt uses only the first 72 bytes and the library throws above that. Sign-in rejects anything over 72 UTF-8 bytes as invalid credentials, and new passwords are limited to 64 characters / 72 bytes
Brute force	5 consecutive failures lock the username for 5 minutes, even for the correct password. A partial run of failures is forgotten after 5 quiet minutes and expired entries are pruned. In memory, per application instance; both numbers are configurable (§6.3)
Password policy	8–64 characters with letters and digits. Passwords are changed through the School Admin's Reset password dialog on the Users & Roles screen. A self-service change exists in the service layer (<code>AuthService.changePassword</code> , same policy) but has no screen yet (§8)
Secrets in memory	Passwords handled as <code>char[]</code> and wiped after hashing/verifying (the JavaFX password field itself keeps a <code>String</code> while the screen is open, a platform limitation)
Session	See §2.7.1: in-memory session, cleared on sign-out and after 15 minutes of inactivity
Log injection	Usernames are sanitised (control characters replaced, length capped) before they reach the log

Measure	Implementation
RBAC	<code>RoleType</code> → <code>Permission</code> matrix (§5.1). Every service method calls <code>session.require(permission)</code> ; the UI additionally hides what a role cannot do
Accountability	<code>SUPERVISION_AUDIT.EXECUTED_BY</code> is taken from the server-side session, and its timestamp is set by a trigger, never by the client
SQL injection	<code>PreparedStatement</code> / <code>CallableStatement</code> with bind variables everywhere
Least privilege (DB)	Application schema user with only <code>CREATE SESSION, TABLE, VIEW, PROCEDURE, TRIGGER, SEQUENCE</code>
Configuration	The shipped jar contains no password (a unit test checks this); credentials live in a git-ignored file or environment variables

2.7.1 Session model: why there is no JWT

- **What exists.** After a successful sign-in `AuthService` stores an immutable `AuthenticatedUser` (id, username, full name, role, linked teacher, sign-in time; never the password hash) in the `UserSession` singleton, an `AtomicReference` so background tasks read the same value as the UI thread. It lives only in the memory of the running JVM: nothing is written to disk, the registry or a cookie, and closing the application ends the session.
- **Why no token.** A token such as a JWT is a signed claim that a *server* checks on every request. This is a two-tier desktop application: the services and the session run in the same process, so there is no second party to present a token to. A JWT here would add a signing key to protect and a token to store, and would protect nothing.
- **If a middle tier is added** (REST API for a web or mobile client), the right design is a short-lived access token (about 15 minutes) kept in memory only, plus a rotating refresh token kept server-side or in the operating-system credential store (Windows Credential Manager, macOS Keychain), never in a plain file. The API would then enforce RBAC with the token's claims.
- **Clean end of session.** Sign-out and the idle timeout (`security.session.idleTimeoutMinutes`, default 15, 0 = off) go through the same path: every open dialog and popup is closed, `logout()` sets the session to empty, the shell stops its timer and detaches its listeners, and the sign-in screen is shown (after a timeout with a notice explaining why). Any later service call fails with `NOT_AUTHENTICATED`. Keyboard, mouse and scroll input in any window of the application counts as activity.

- **Known boundary.** RBAC is enforced in the Java service layer, and all clients connect with one schema account. Someone who has the configuration password could bypass RBAC with SQL*Plus, although every business rule still holds because it lives in PL/SQL, constraints and triggers. Closing that gap needs per-user database identities (Oracle proxy authentication or secure application roles) or a middle tier that alone holds the credentials.

2.8 Internationalisation and right-to-left

- **Text.** Two UTF-8 bundles hold all text; FXML uses `%key`, code uses `Messages.get(key, args)`. The language is chosen on the login screen and remembered per OS user (Java Preferences).
- **Layout.** In Arabic the scene root gets `NodeOrientation.RIGHT_TO_LEFT`, which mirrors all layout, tables and dialogs.
- **Fonts.** Arabic uses Tahoma or Geeza Pro, because JavaFX's fallback font drops spaces between Arabic words and digits.
- **Numbers.** Arabic dates use Arabic-Indic digits. Times and fractions are written without spaces (`09:00–11:00`, `16/24`) so the bidi algorithm keeps them as one left-to-right run, in the UI and in PDFs.
- **Reports.** PDFs are built entirely from RTL table cells (OpenPDF only shapes Arabic there). Excel sheets are set right-to-left.

3. Database and business-logic engine

The schema (owner `EXAM_ADMIN`) has 13 tables in four groups:

Group	Tables	Changes
Security	<code>ROLES</code> , <code>USERS</code>	rarely; School Admin only
Organisation	<code>DEPARTMENTS</code> , <code>TEACHERS</code> , <code>COURSES</code> , <code>STUDENTS</code> , <code>TEACHER_STUDENT_RELATIONS</code>	master data, maintained before each term
Infrastructure	<code>ROOMS</code> , <code>EXAM_PERIODS</code>	master data
Operations	<code>EXAM_SCHEDULE</code> , <code>SEATING_ALLOCATION</code> , <code>SUPERVISION_ROSTER</code> , <code>SUPERVISION_AUDIT</code>	written by the PL/SQL engine during exam preparation and on exam day

Only the operations group is written by the business-logic engine; the other groups are plain master data behind validation, constraints and triggers.

3.1 Entity–relationship model

Tables and relationships (columns and constraints are listed in §3.2).

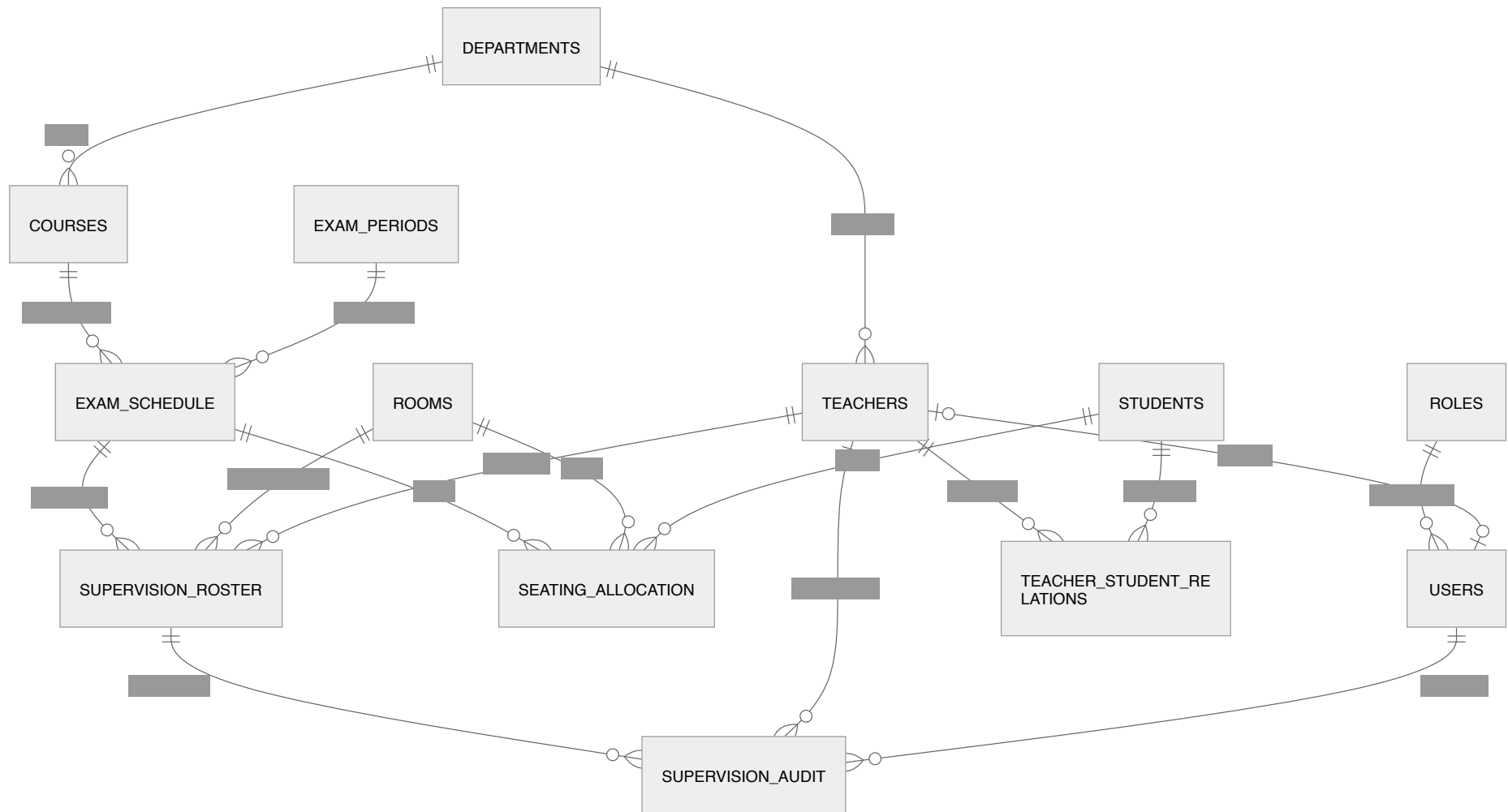


Figure 3 - Entity-relationship model

3.2 Table catalogue

All primary keys are `GENERATED BY DEFAULT ON NULL AS IDENTITY`. Every constraint is named, so errors can be translated precisely.

#	Table	Purpose	Notable rules
1	ROLES	SCHOOL_ADMIN , CONTROL_OFFICER , COMMITTEE_HEAD , TEACHER	names match the Java RoleType enum
2	USERS	Logins	BCrypt format CHECK; soft delete; optional unique TEACHER_ID
3	DEPARTMENTS	Academic departments	drive the subject- conflict rule
4	TEACHERS	Proctor pool	MAX_WEEKLY_LOAD ≥ MAX_DAILY_LOAD , daily 0–10, HOURS_BALANCE ≥ 0 ; soft delete
5	COURSES	Course → department + grade level	unique code
6	ROOMS	Halls	0 < EXAM_CAPACITY ≤ REGULAR_CAPACITY ; status AVAILABLE / MAINTENANCE / UNAVAILABLE (shown as “Suspended”); soft delete
7	EXAM_PERIODS	Time-of-day templates (“First Period” 09:00–11:00)	only the time part of the DATEs is used
8	EXAM_SCHEDULE	An exam = course + date + period	one exam per course per day; date without time
9	STUDENTS	Roster	grade level, section, HAS_SPECIAL_NEEDS
10	SEATING_ALLOCATION	Student → room/seat for an exam	one seat per student per exam; unique seat

#	Table	Purpose	Notable rules
			per exam+room; attendance status; cascade-deleted with its exam
11	SUPERVISION_ROSTER	Teacher → room for an exam, role head/proctor	status <code>CONFIRMED</code> / <code>REPLACED</code> / <code>CANCELLED</code> ; function- based unique indexes: one <i>active</i> assignment per teacher per exam, one <i>active</i> head per exam+room
12	SUPERVISION_AUDIT	Emergency substitution log	append-only (trigger), server timestamp
13	TEACHER_STUDENT_RELATIONS	Kinship, degree 1–4	unique teacher+student pair

Schema extensions agreed during development (on top of the original 13-table specification):

Extension	Reason
<code>UPDATED_AT</code> , <code>DELETED_AT</code> on <code>USERS</code> , <code>TEACHERS</code> , <code>ROOMS</code>	Audit timestamps maintained by triggers
<code>USERS.TEACHER_ID</code> (nullable, unique, FK)	Teacher and Committee Head logins see their own duties
<code>STUDENTS.SECTION</code> (nullable)	Class section shown and edited in the student roster

Views.

View	Purpose
V_USERS , V_TEACHERS , V_ROOMS	Active rows only. The DAOs read from these and delete through them
V_EXAM_SLOTS	Every exam with its real SLOT_START / SLOT_END (exam date + period time) and duration in hours; the basis of all overlap logic
V_SUPERVISION_ROSTER	Active roster with names, for reports

3.3 Soft delete

Oracle cannot turn a `DELETE` on a *table* into an `UPDATE`. The design therefore uses views:

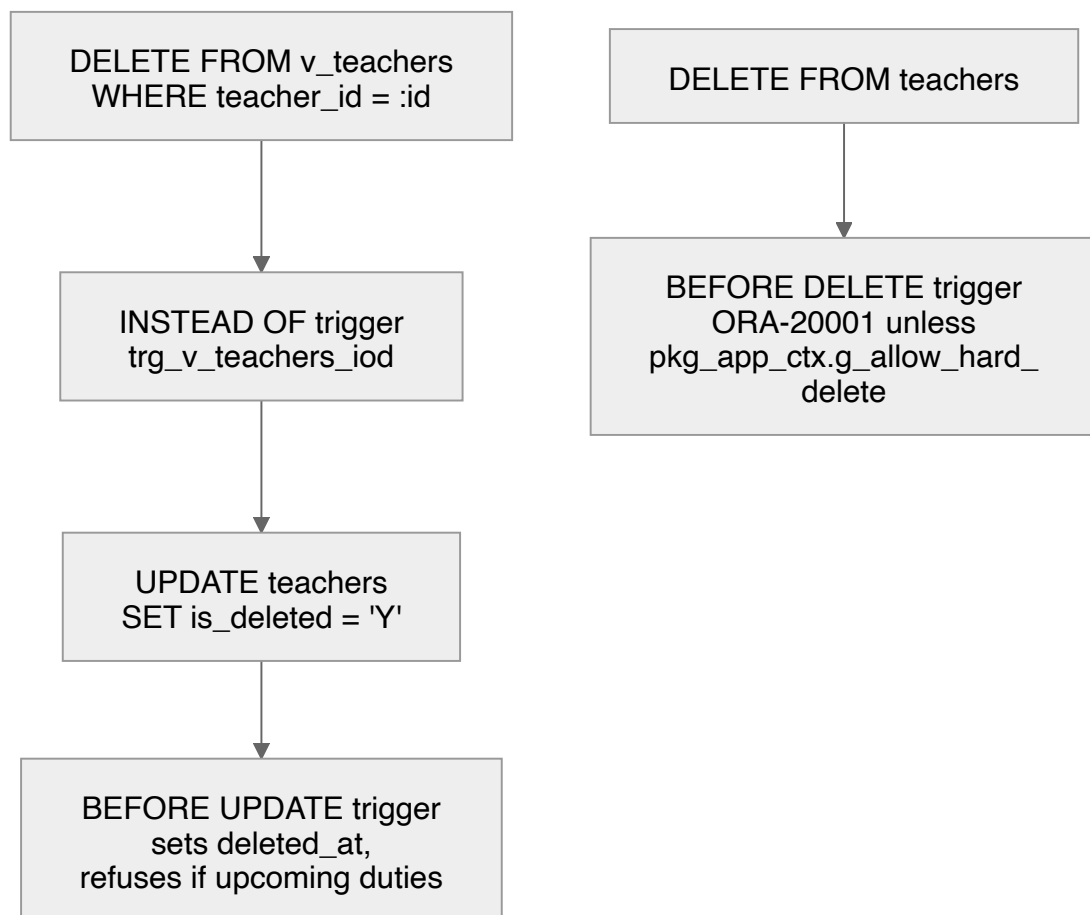


Figure 4 - Soft delete through views

- **Restoring a row** is `UPDATE ... SET is_deleted = 'N'`; the trigger clears `DELETED_AT`.

- **History survives archiving.** Archived teachers still appear in historical reports: the compensation report reads the base table, because an archived teacher must still be paid for work done.

3.4 Triggers (16)

Trigger	Event	Effect
trg_users_bi / _bu	insert / update	trim username; CREATED_AT immutable; UPDATED_AT , DELETED_AT
trg_users_bd , trg_teachers_bd , trg_rooms_bd	delete on base table	block hard delete (ORA-20001)
trg_v_users_iod , trg_v_teachers_iod , trg_v_rooms_iod	INSTEAD OF delete on view	soft delete
trg_teachers_bi / _bu	insert / update	upper-case code; refuse archiving with upcoming CONFIRMED duties (ORA- 20002)
trg_rooms_bi / _bu	insert / update	upper-case code; refuse archiving, leaving AVAILABLE, or shrinking exam capacity while future exams use the room (ORA-20002)
trg_seating_biu	insert / room change	seat only in active AVAILABLE rooms (ORA-20004)
trg_roster_biu	insert / teacher or room change	assign only active teachers to usable rooms (ORA-20005 / 20004)
trg_audit_bi	insert	AUDIT_TIMESTAMP := SYSTIMESTAMP (client value ignored)
trg_audit_bud	update / delete	audit is append-only (ORA-20003)

3.5 Audit logging

The audit trail has three layers:

1. **SUPERVISION_AUDIT**. Every emergency substitution records the original roster row (which keeps the replaced teacher), the substitute, the executing user, the reason and a server timestamp. Rows cannot be updated or deleted.
2. **Roster history**. Replaced rows stay in **SUPERVISION_ROSTER** with status **REPLACED**, and cancelled allocations with **CANCELLED**. Nothing is overwritten.
3. **Application log** (**logs/examhalls.log**, rolling daily, 14 days). It records who signed in, generated seating, allocated, substituted, exported, and created or edited master data.

3.6 PL/SQL packages

Package	Routine	Contract
pkg_app_ctx (spec only)	error-code constants, g_allow_hard_delete	shared by triggers and packages
pkg_exam_util	get_slot(exam_id)	the exam's date, real start/end, duration, course department, grade
pkg_seating	generate_seating(exam_id, OUT seated, OUT rooms_used)	§4.2
	clear_seating(exam_id)	refused while proctors are allocated
	mark_attendance(seating_id, status)	PRESENT / ABSENT / EXCUSED / ABSENT_PENDING
pkg_supervision	check_eligibility(teacher, exam, room) → NULL or 'CODE: reason'	§4.4
	allocate_proctors(exam, OUT assigned, OUT unfilled, students_per_proctor = 20, allow_partial = 'N')	§4.3
	cancel_allocation(exam, OUT cancelled)	releases staff and refunds hours
	find_best_substitute(roster_id) → teacher_id	read-only preview
	replace_proctor(roster_id, executed_by, reason, IN OUT substitute_id, OUT new_roster_id, OUT audit_id)	§4.5
	get_candidates(exam, room) → SYS_REFCURSOR	every active teacher with an eligibility column

Transaction contract. Procedures never `COMMIT`. Each sets a `SAVEPOINT` and rolls back to it on any error, so a failed call leaves no partial work. The Java service decides when to commit, and calls use named notation (`p_exam_id => ?`) so a reordered package spec cannot break them.

Concurrency.

- **Seating:** locks the exam row and its period row (`FOR UPDATE WAIT 10`), so two users cannot grab the same free seats.
- **Allocation:** picks a teacher, locks the teacher row, then **re-checks** eligibility before inserting (“pick, lock, re-check”), because another session may have assigned the same teacher meanwhile.
- **Last line of defence:** the function-based unique indexes on the roster.

3.7 Application error codes

Code	Name	Raised when
-20001	HARD_DELETE_BLOCKED	<code>DELETE</code> on USERS / TEACHERS / ROOMS base table
-20002	HAS_FUTURE_DUTIES	archiving a busy teacher, or taking a room in use out of service
-20003	AUDIT_IMMUTABLE	updating/deleting <code>SUPERVISION_AUDIT</code>
-20004	ROOM_NOT_USABLE	seating/assigning in a room that is not AVAILABLE
-20005	TEACHER_NOT_ACTIVE	assigning an archived teacher
-20010	NOT_FOUND	unknown exam / roster row / user
-20011	INSUFFICIENT_CAPACITY	students do not fit into the free seats of the slot
-20012	STUDENT_CLASH	students already seated in an overlapping exam
-20013	NO_STUDENTS	no students in the course's grade level
-20020	NO_ELIGIBLE_TEACHER	allocation or substitution cannot fill a position
-20021	TEACHER_INELIGIBLE	chosen substitute violates a rule
-20022	INVALID_STATE	e.g. re-seating a staffed exam, replacing a non-CONFIRMED row
-20030	INVALID_ARGUMENT	bad parameter (e.g. empty reason)

4. Algorithms and fairness rules

4.1 Time model

An exam's real slot is `EXAM_DATE + time-of-day(PERIOD)`. Two exams **overlap** when

```
a.slot_start < b.slot_end AND b.slot_start < a.slot_end
```

Every rule (room capacity, student clashes, teacher availability, relatives) uses this test rather than “same period id”, so it stays correct if periods are redefined or several exams share a slot.

4.2 Seating generator (`pkg_seating.generate_seating`)

```

1  lock the exam row and its period row
2  refuse if the exam has active proctors
   (cancel the allocation first)
3  delete the exam's existing seating (safe to regenerate)
4  S := students of the course's grade level,
   ordered by has_special_needs DESC, student_code
5  refuse if any of S sits an overlapping exam
   (STUDENT_CLASH)
6  refuse if a teacher already supervising the slot
   is related to any of S
7  R := active AVAILABLE rooms with free seats > 0, where
   free = exam_capacity - seats used by overlapping exams
   ordered by free DESC, room_code (largest first)
8  for each room r in R while students remain:
   take min(free(r), remaining) students
   number seats after the highest seat already used
   in r during the slot ("001", "002", ...)
9  if students remain: rollback, INSUFFICIENT_CAPACITY
   (all-or-nothing)
10 bulk insert (FORALL)
```

- **Largest rooms first** minimise the number of rooms, and so the number of proctors needed.
- **Special-needs students first** get the lowest seat numbers, i.e. the front rows of the largest room.

- **Room sharing.** A second exam in the same slot uses the remaining capacity of partly filled rooms, and its seat numbers continue after the first exam's, so a physical seat is never double-booked.
- **Cost:** after sorting, the work is linear in students + rooms, with a single bulk insert.

Seed example (10 Jan 2027, First Period). MATH-10 (70 students) is seated in HALL-1 (60) and B101 (10). CHEM-11 (70 students, same slot) then takes the rooms with the most free seats left: B102 (30), A101 (24) and A102 (16). With less capacity it would continue into B101's 20 remaining seats, numbering them 011 onwards.

4.3 Fair proctor allocation (`pkg_supervision.allocate_proctors`)

Staffing rule, per room used by the exam:

```
students_in_room = students seated in the room during the slot
                  (all exams sharing the room)
heads_needed     = 1
proctors_needed  = max(1, ceil(students_in_room / 20))
missing          = needed - staff already in the room during the slot
```

Staff attached to another exam that shares the room are counted, so a shared room is never double-staffed.

Fairness rule. Each open position is given to the first eligible teacher in this order:

```
ORDER BY hours_balance ASC,      -- least supervision hours
       active CONFIRMED duties ASC, -- then fewest current duties
       teacher_id                -- deterministic tie-break
```

`HOURS_BALANCE` grows by the slot's duration on each assignment, decreases on cancellation, and moves from the replaced teacher to the substitute. The ordering is re-evaluated after every single assignment inside the same transaction, so load spreads evenly across the whole day and week.

`pick_teacher` walks the candidates in that order and takes the first one for which `check_eligibility` returns no reason. It then locks that teacher's row, re-checks, inserts the CONFIRMED roster row and adds the slot duration to `HOURS_BALANCE`.

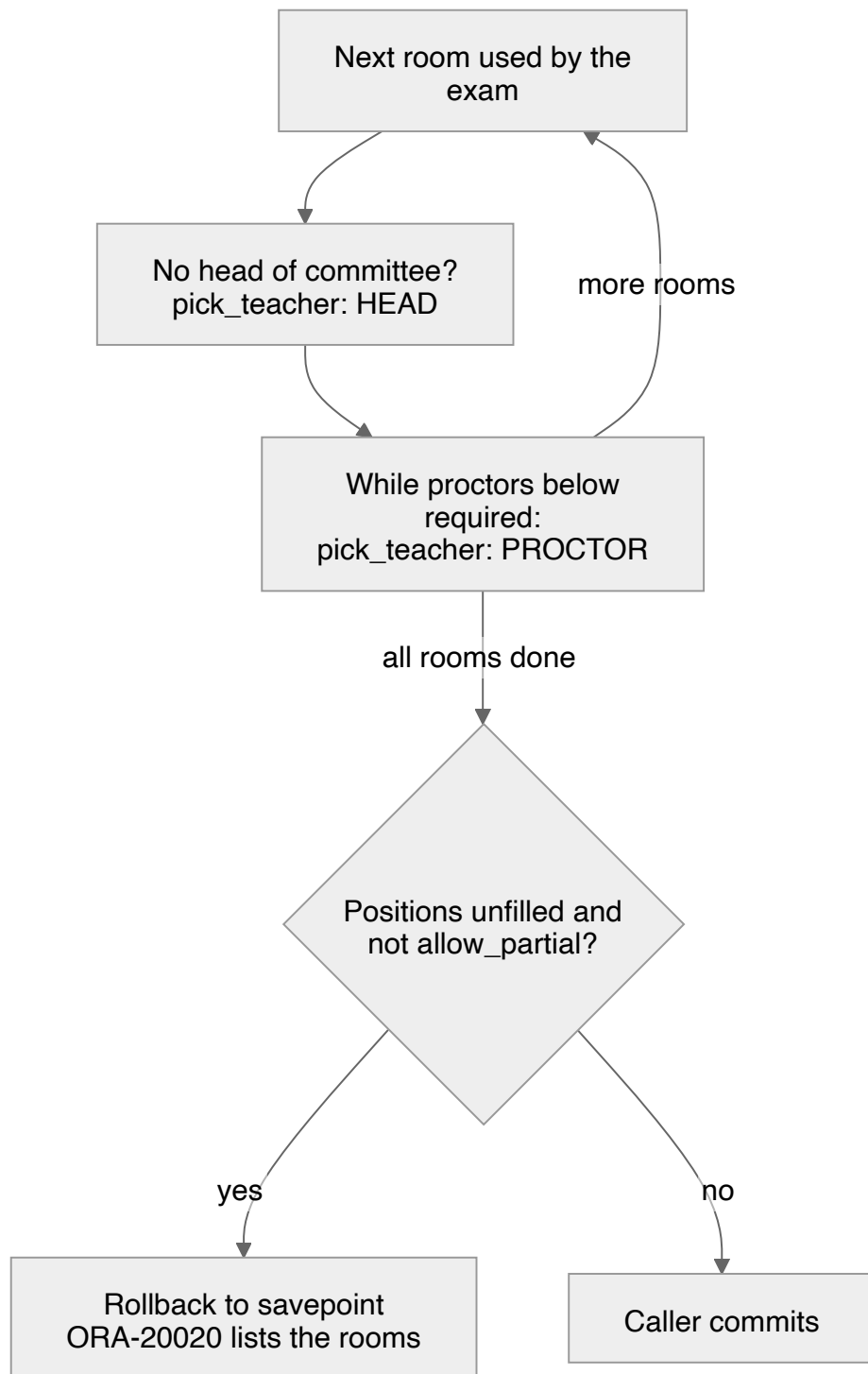


Figure 5 - Fair proctor allocation

4.4 Conflict-of-interest and eligibility rules

(`check_eligibility`)

The checks run in this order; the first failing one is returned as `CODE: explanation` and shown translated in the UI.

#	Code	Rule	Scope
0	NOT_FOUND / TEACHER_INACTIVE	teacher exists and is not archived	—
1	SUBJECT_CONFLICT	the teacher's department equals the department of the exam's course, or of any other course seated in the same room during the slot	per room: a Chemistry teacher may supervise a Maths-only room during a shared slot
2	TIME_CONFLICT	already supervising an overlapping slot	teacher
3	DAILY_LIMIT	confirmed sessions that day $\geq$ MAX_DAILY_LOAD	teacher, day
4	WEEKLY_LIMIT	confirmed sessions in that ISO week $\geq$ MAX_WEEKLY_LOAD	teacher, Monday-based week
5	RELATIVE_IN_SESSION	a relative (degree 1–4) sits any exam, in any room, during an overlapping slot — or belongs to the grade of this exam (even before it is seated)	whole session (strict exam-control standard)

The relatives rule is also enforced in the other direction:

- **Seating** refuses to seat a grade when a teacher already supervising that slot is related to one of its students.
- **Recording a new relation** reports how many of the teacher's upcoming duties now conflict, so the Control Officer can substitute them.

4.5 Emergency 1-click substitution

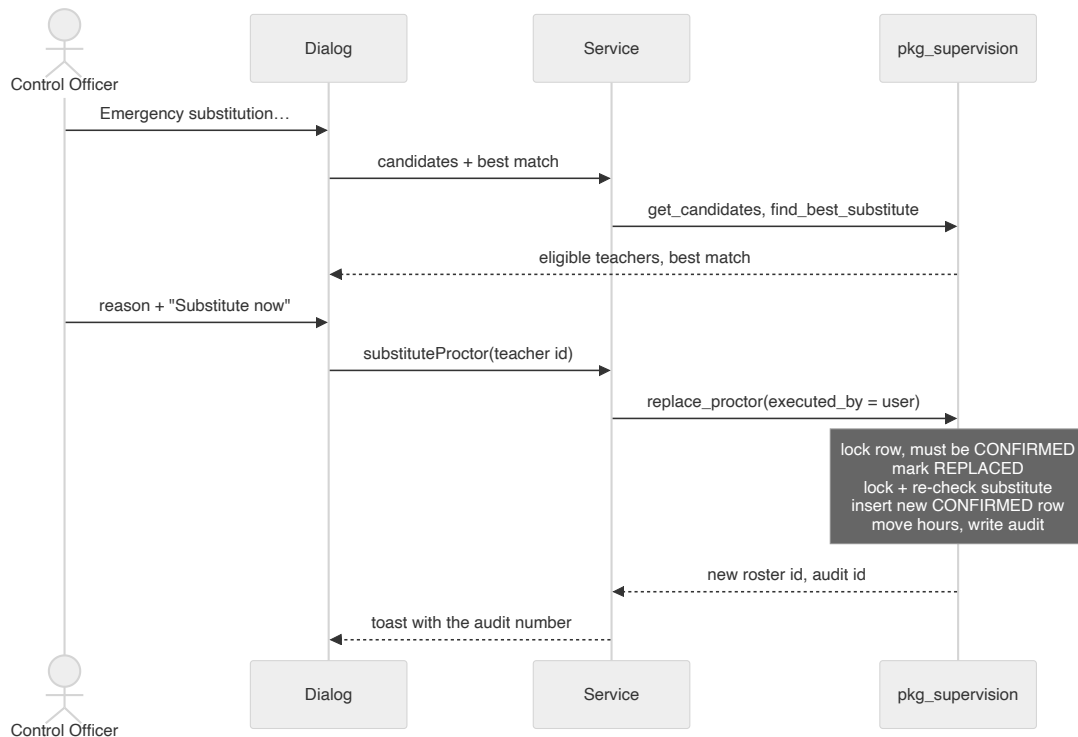


Figure 6 - Emergency 1-click substitution

- **What you see is what is assigned.** The dialog sends the *explicit* id of the teacher it displayed. If that teacher became ineligible in the meantime, the call fails with a clear reason instead of silently picking someone else.
- **"Best match"** uses the same fairness ordering as allocation, excluding the replaced teacher.
- **Manual choice.** The candidate list shows only eligible teachers; a toggle shows everyone with the translated reason they are blocked.

4.6 Under-staffing detection (Occupancy Grid, daily control sheet)

For every room × period of a day:

```

EMPTY          no seated students, room AVAILABLE
UNAVAILABLE    room under maintenance / suspended, empty
UNDERSTAFFED   seated > 0 and (no head, or
                proctors < max(1, ceil(seated / 20)))
STAFFED        otherwise

missing = (no head ? 1 : 0)
          + max(0, required proctors - proctors)

```

4.7 Dashboard metrics

The Executive Dashboard reads everything with six read-only queries (exam overview, rooms, seats, roster, department workload, audit feed) and computes the figures in memory for all exams from today on. Staffing is measured per **room session** (room + day + period), the same unit the allocation engine uses, because exams sharing a room share its staff.

```

required(session) = 1 head + max(1, ceil(seated / 20))
filled(session)   = min(1, heads) + min(proctors, required proctors)
coverage          = sum(filled) / sum(required)
capacity(day)     = sum(exam capacity of AVAILABLE rooms)
                  x periods used that day
red alert         = seated session with nobody on duty
amber alert       = seated session partly staffed

```

Each exam gets a state for the donut chart: **awaiting seating** (nothing seated), **unassigned** (seated, nobody on duty in any of its rooms), **partially staffed** or **fully staffed**. The focus day is today when it has exams, otherwise the next exam day. It drives the exam table, the teachers-on-duty figure and the quick actions. The substitution feed is shown only to roles with `VIEW_AUDIT`.

4.8 Compensation hours

The Excel report sums the duration of **CONFIRMED** duties in the chosen date range. A substitute is paid for the session; the replaced teacher is not. The hourly rate is an editable cell (default from `reports.compensation.hourlyRate`), and the amount and total columns are live Excel formulas.

5. Role-based user guide

5.1 Permission matrix

Permission	School Admin	Control Officer	Committee Head	Teacher
Manage users & roles	✓			
Manage master data (teachers, rooms, courses, students, exam schedule)	✓	✓		
Generate seating	✓	✓		
Allocate / cancel proctors	✓	✓		
Emergency substitution	✓	✓		
View all schedules, grid, seating & supervision	✓	✓	✓	
Reports & exports	✓	✓	✓	own schedule only
View audit trail	✓	✓		
View own duties	✓	✓	✓	✓

Landing page after sign-in: Admin and Control Officer → Exam Schedule; Committee Head → Dashboard; Teacher → My Duties. The sidebar only lists pages the role may open, and the services re-check every permission.

5.2 Seed accounts (change the passwords after the first sign-in)

Username	Password	Role	Linked teacher
admin	Admin@2026	School Admin	—
control	Control@2026	Control Officer	—
head	Head@2026	Committee Head	T-PHYS-01 Omar Tarek Mansour
teacher	Teacher@2026	Teacher	T-ENGL-01 Sarah Kamal Wahba

5.3 Common to everyone

- **Sign-in.** Enter username and password. The **English / العربية** switch under the button changes the language for every screen and report, and the choice is remembered.
- **Status bar.** Shows whether the database is reachable. Successful actions show a green confirmation in the top bar, and errors explain what to do, in the current language.
- **Sign out.** Bottom of the sidebar. After 15 minutes without keyboard or mouse activity the application signs out by itself, closes any open dialog and explains why on the sign-in screen.
- **Dashboard** (Admin, Control Officer, Committee Head). Five KPI cards: active exams, seated students vs hall capacity, proctoring coverage, under-staffed rooms (red/amber) and teachers on duty. Three charts: capacity vs seated per exam day (click a day to open its Occupancy Grid), proctor allocation status and department workload. Below them, the exams of the focus day with their staffing status (double-click to open one), quick actions (Seating & Supervision, Occupancy Grid, daily control sheet PDF) and the latest emergency substitutions. Hover over a card or bar for the exact definition and figures; **Refresh** reloads everything.

5.4 School Admin

1. **Users & Roles.** Create accounts (username 3–50 characters, strong password), assign the role, and link Teacher / Committee Head accounts to their teacher record (required for Teacher). Other actions:

- reset a password (**Reset password** sets a new one under the same policy; this is currently the only way to change a password, as there is no self-service screen yet, see §8);
- delete (archive) an account, which keeps its history.

The system refuses to delete your own account or to remove the last School Admin.

2. Everything the Control Officer can do (below).

5.5 Control Officer — preparing an exam session

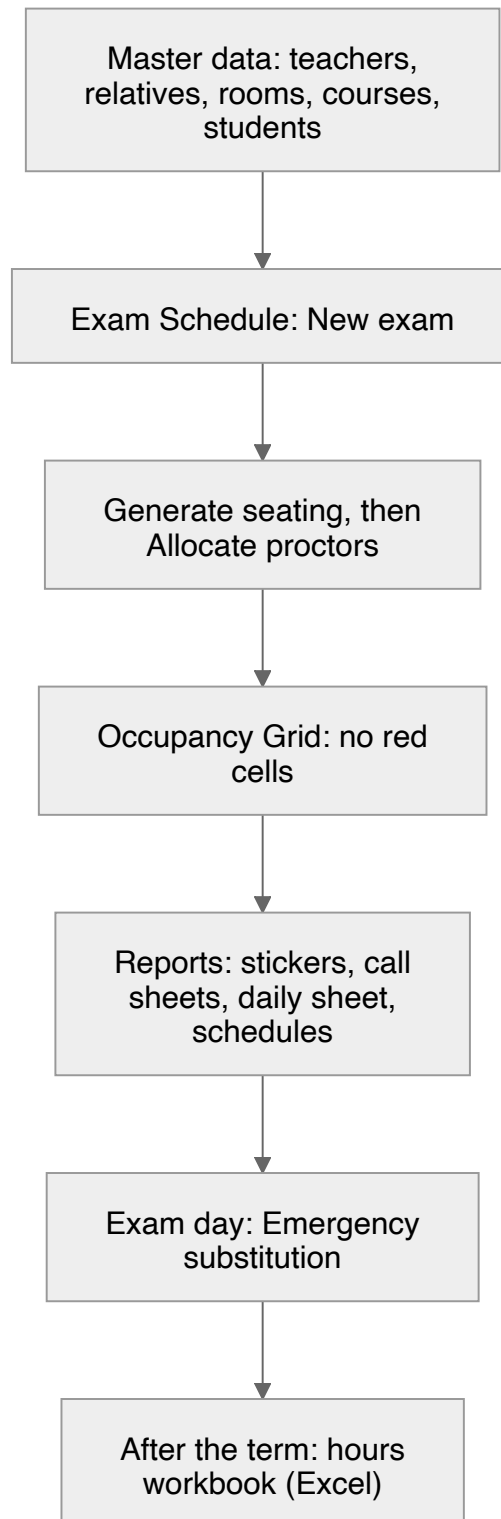


Figure 7 - Control Officer workflow

1. **Master data** (Management section of the sidebar):
 - **Teachers & Staff:** employee code, name, department (drives the subject rule), daily and weekly caps; the hours balance is read-only. Select a teacher to see and record **relatives** (degree 1–4). Archiving is refused while the teacher has upcoming duties.
 - **Rooms & Halls:** regular vs exam capacity (the table shows the spacing ratio) and status. Suspending a room, shrinking its capacity or archiving it is refused while upcoming exams use it.
 - **Courses & Departments:** each course belongs to a department and a grade level.
 - **Students:** search, filter by grade or “special needs only”, edit section and special-needs flag.
2. **Exam Schedule** → **New exam:** course, period, date, notes (e.g. “Calculators not allowed”). A course can be examined once per day. A seated exam can only have its notes edited until its seating is cleared.
3. **Select the exam** → **Generate seating.** The status becomes **Seated**. Schedule every exam of a time slot before allocating, so shared rooms and relatives are handled correctly.
4. **Allocate proctors.** The status becomes **Staffed**. If a position cannot be filled, nothing is saved and the message lists the rooms concerned.
5. **Seating & Supervision** (Open, or double-click the exam):
 - the supervision matrix shows each room’s occupancy, head of committee and proctors;
 - the lower tabs show the room’s staff and seating plan;
 - **Cancel allocation** releases the staff and refunds their hours.
6. **Occupancy Grid:** check the day. Red cells are under-staffed; click a cell to open that exam.
7. **Reports & Exports:** seating stickers and attendance sheets per exam, the daily control sheet per day, and individual schedules (one teacher or all). Files open automatically after saving.
8. **Exam day — a proctor is absent:**
 1. Open Seating & Supervision, select the room and the staff row, then **Emergency substitution....**
 2. Check the best match, choose or type a reason, and click **Substitute now** (or pick another eligible teacher).
 3. The change and the audit record are saved immediately.

9. **After the term:** Reports → *Proctoring hours for compensation* (Excel); enter the hourly rate in the yellow cell.

5.6 Committee Head

- **Dashboard:** your landing page: KPIs, charts, the exams of the day and quick actions (the substitution feed is reserved for the control office).
- **Exam Schedule / Occupancy Grid / Seating & Supervision:** read-only; see the rooms, students and staff of every session.
- **Reports:** print the attendance call sheet for your room and the daily control sheet for signatures.
- **My Duties:** your own upcoming duties, with **Download PDF**.

5.7 Teacher

- **My Duties:** date, period, course, room and role of every upcoming supervision.
- **Download PDF:** your individual proctoring schedule. Teachers can only export their own schedule.

6. Deployment and setup guide

6.1 Requirements

Component	Requirement
Client PC	Windows 10/11 x64, Linux x64 or macOS; 1280×720 or larger screen. Portable Windows edition: nothing to install. Other bundles: Java 21 or newer (e.g. Eclipse Temurin 21)
Database	Oracle Database 19c or newer, or Docker Desktop to run Oracle Database Free (≈ 2 GB RAM, 1.5 GB download)
Build machine (only to build from source)	JDK 21+, Maven 3.9+
Fonts for Arabic PDFs	Tahoma or Arial (present on Windows and macOS); on Linux DejaVu Sans or Noto Sans Arabic

6.2 Oracle Database Free in Docker

```
docker run -d --name examhalls-oracle -p 1521:1521 \
  -e ORACLE_PASSWORD=<sys-password> \
  -e APP_USER=EXAM_ADMIN -e APP_USER_PASSWORD=<app-password> \
  -v examhalls-oradata:/opt/oracle/oradata \
  gvenzl/oracle-free:23-slim

# wait until the log says: DATABASE IS READY TO USE!
docker logs -f examhalls-oracle
```

The image creates the `EXAM_ADMIN` schema in the pluggable database `FREEPDB1`. Data persists in the `examhalls-oradata` volume. After a reboot, run `docker start examhalls-oracle`.

Install schema, packages and seed data (drops and recreates all application objects, then runs the smoke test):

```
# macOS / Linux
APP_USER_PASSWORD=<app-password> database/install-docker.sh
```

```
rem Windows
set APP_USER_PASSWORD=<app-password>
database\install-docker.bat
```

Existing Oracle server instead of Docker. A DBA creates the user (see `database/README.md`); then run `database/install.sql` from that folder with SQL*Plus, SQLcl or SQL Developer (*Run Script*, F5):

```
CREATE USER exam_admin IDENTIFIED BY "<password>"
        QUOTA UNLIMITED ON users;
GRANT CREATE SESSION, CREATE TABLE, CREATE VIEW,
        CREATE PROCEDURE, CREATE TRIGGER, CREATE SEQUENCE TO exam_admin;
```

6.3 Application configuration

Copy `config/application.properties.example` to `config/application.properties` next to the application (or the project root when running from source):

```
# format: jdbc:oracle:thin:@//host:port/service
db.url=jdbc:oracle:thin:@//localhost:1521/FREEPDB1
db.username=EXAM_ADMIN
db.password=<app-password>
```

Key	Default	Meaning
<code>db.url</code> , <code>db.username</code> , <code>db.password</code>	localhost / EXAM_ADMIN / empty	connection (or <code>DB_URL</code> , <code>DB_USERNAME</code> , <code>DB_PASSWORD</code> environment variables)
<code>db.pool.maximumPoolSize</code> / <code>minimumIdle</code>	10 / 2	pool size
<code>db.pool.connectionTimeoutMs</code>	5000	wait for a connection
<code>db.pool.initializationFailTimeoutMs</code>	1	one login attempt at pool start (keep > 0)
<code>db.oracle.connectTimeoutMs</code>	4000	network connect timeout
<code>app.school.name</code>	Exam Control Office	printed in every report header
<code>reports.font.regular</code> / <code>reports.font.bold</code>	auto-detect	TrueType font with Arabic glyphs for PDFs
<code>reports.compensation.hourlyRate</code>	50	default rate in the hours workbook
<code>security.login.maxAttempts</code> / <code>security.login.lockMinutes</code>	5 / 5	failed sign-ins before a username is locked, and for how long
<code>security.session.idleTimeoutMinutes</code>	15	automatic sign-out after inactivity (0 = never)

The file contains a password: restrict its permissions on shared lab PCs, or use the `DB_PASSWORD` environment variable instead.

6.4 Build and run from source

```
mvn javafx:run           # run in development
mvn test                 # 84 test cases (integration suites
expect fresh seed data)
mvn -Pdist clean package # lab bundle
```

`mvn -Pdist clean package` produces:

- `target/exam-halls-proctoring-1.0.0-all.jar` — the runnable fat jar (all libraries, views, styles, translations);
- `target/exam-halls-proctoring-1.0.0-dist.zip` — the lab bundle:

```
exam-halls-proctoring-1.0.0/
├── examhalls.jar          app + JavaFX natives (Win, Linux, macOS)
├── run-app.bat            Windows launcher
├── run-app.sh             macOS / Linux launcher
├── config/application.properties.example
├── database/              00-05 scripts, install.sql,
                           install-docker.sh / .bat
├── README.md
└── DOCUMENTATION.md
```

The default (non-`dist`) jar only contains JavaFX natives for the machine that built it. The `dist` profile adds the Windows (`.dll`, Direct3D) and Linux (`.so`, GTK) natives, so one bundle built on any machine runs on the Windows lab PCs.

Portable Windows edition (no Java installation needed).

```
# runs mvn -Pdist, then adds the Java runtime
packaging/build-windows-portable.sh
```

This produces `target/exam-halls-proctoring-1.0.0-windows-x64-portable.zip` (≈ 83 MB). It is the lab bundle above plus:

- `runtime\` — the official **Eclipse Temurin 21 JRE for Windows x64** (Java 21.0.12). It is downloaded from the Adoptium API into `target/jre-cache/`, verified against Adoptium's published SHA-256, and cached for later builds.
- `PORTABLE-EDITION.txt` — three-step instructions for the lab.
- `DOCUMENTATION.pdf` — this document, regenerated from `DOCUMENTATION.md` with `packaging/build-docs-pdf.sh` (also included in the regular lab bundle).

`run-app.bat` / `run-app.sh` look for Java in this order: `runtime\` next to the script, then `JAVA_HOME`, then the `PATH`. With the portable edition no Java installation or environment variable is needed.

Why the full JRE instead of `jlink`: `jlink` can only link runtime modules of its own Java version, so a trimmed Windows Java 21 runtime needs two extra full JDKs (~400 MB) at build time. A module forgotten by mistake would also only fail on the lab PCs. The full JRE (70 modules) costs a few tens of MB more and removes that risk.

6.5 Running on a lab machine

1. Unzip the bundle; with the **portable edition** skip Java entirely, otherwise install Java 21+.
2. Create `config\application.properties` (§6.3) pointing to the database server.
3. Double-click `run-app.bat` (or run `./run-app.sh`). The launcher:
 - checks that Java 21+ is installed and finds `examhalls.jar`;
 - warns if the configuration file is missing;
 - runs `java -jar examhalls.jar --check-db` — **one** login attempt that prints `[OK] Database reachable...` or `[FAILED] <reason>` in a few seconds;
 - on failure lists the usual fixes and asks *Start the application anyway? [Y/N]*;
 - starts the application with `-Dfile.encoding=UTF-8` (Windows: `javaw`, so no console window stays open).

Options: `run-app.bat --no-check` skips the database check, and `JAVA_OPTS` overrides the memory settings (default `-Xms128m -Xmx768m`).
4. Logs are written to `logs\examhalls.log` next to the application.

6.6 Troubleshooting

Symptom	Cause / fix
[ERROR] Java 21 or newer is required	Install Java 21+ or set <code>JAVA_HOME</code>
[FAILED] Cannot reach the database ... ORA-12541	Container or listener not running: <code>docker start examhalls-oracle</code> , check host/port in <code>db.url</code> , firewall on 1521
[FAILED] The database rejected the configured account ... ORA-01017	Wrong <code>db.username</code> / <code>db.password</code>
ORA-28000 account locked	As SYSTEM: <code>ALTER USER exam_admin ACCOUNT UNLOCK;</code> (fix the password first)
PDF export: "No font with Arabic support"	Set <code>reports.font.regular</code> / <code>reports.font.bold</code> to a <code>.ttf</code> with Arabic glyphs
Console shows <code>Unsupported JavaFX configuration: classes were loaded from 'unnamed module'</code>	Harmless notice when JavaFX runs from a classpath jar
Arabic: a few labels join an Arabic word to the following number on macOS	JavaFX text-rendering quirk with the macOS fallback font (see §8)
Integration tests fail after using the app	They expect fresh seed data: reinstall with <code>install-docker</code>
Reset the demo data	Run <code>install-docker</code> again (drops and recreates everything)

7. Quality assurance

Suite	What it proves	Count
<code>OracleErrorTranslatorTest</code>	ORA codes → ErrorCode, constraint-specific messages, login vs network errors, cause-chain unwrapping	8
<code>MessagesCompletenessTest</code>	identical key sets in EN/AR, every ErrorCode and role translated, Arabic really Arabic, argument formatting	4
<code>RoleTypeTest</code> , <code>LoginAttemptTrackerTest</code>	permission matrix; lockout, unlock, forgetting and pruning with a fake clock	5
<code>CredentialPolicyTest</code>	accepted and rejected usernames (SQL injection strings, spaces, quotes, NUL and bidi control characters, Arabic, too long); 72-byte limit; password policy; log sanitiser	25
<code>AuthServiceSecurityTest</code>	with a fake DAO that records every lookup: malformed input never reaches the database, empty/null input, over-long passwords fail cleanly, whitespace stripping, identical errors, lockout even for the correct password, password arrays wiped, logout clears the session, new-password bounds	17
<code>IdleTimerTest</code>	idle timeout expiry, activity reset, disabled at 0	3
<code>ResourceConfigTest</code>	<code>logback.xml</code> is well-formed; the jar ships no password and fails fast	2
<code>DatabaseIntegrationTest</code>	login and RBAC, full exam-day workflow through PL/SQL, stages and supervision matrix, teacher-linked duties, soft delete and constraint translation	8

Suite	What it proves	Count
<code>ReportsAndUsersIntegrationTest</code>	occupancy grid alerts, every PDF/Excel report in both languages, teacher export restriction, user lifecycle and guards	4
<code>MasterDataIntegrationTest</code>	teachers and relatives (conflict reporting), room interlocks, department/course/student rules, exam editor guards, permissions	5
<code>DashboardIntegrationTest</code>	dashboard figures are internally consistent on any data set; role visibility (no feed for Committee Head, no dashboard for Teacher); seating and staffing an exam moves seated, state, coverage and on-duty figures	3
<code>database/05_smoke_test.sql</code>	25 PL/SQL-level checks (capacity, shared rooms, subject, time and relatives rules, substitution, audit immutability, soft delete)	25

- **No trace left behind.** Integration tests run inside `TransactionManager.runAndRollback`, so they never change the data.
- **No database, no failure.** Integration tests are skipped automatically when the database is not reachable, so `mvn package` works anywhere.
- **Checked by hand as well.** During development every screen was also driven end to end in both languages by a scripted JavaFX driver, and the rendered PDFs were inspected visually.
- **Java 21 runtime.** For the portable edition the same driver flows (login, seating, allocation, substitution, reports, users, master data, English and Arabic) were run on an Eclipse Temurin 21 JRE from the fat jar on the classpath, exactly as on the lab PCs; all passed.

8. Known limitations and future work

Area	Status / idea
Attendance	Recorded on the printed call sheets; the service and PL/SQL API (<code>mark_attendance</code>) exist, a screen for live digital attendance is future work
Exam periods	Managed in SQL (seed data); a small period editor would complete master data
Academic terms	Exams are unique per course and date; an explicit term/year dimension would allow archiving by term
Audit viewer	Substitutions are in <code>SUPERVISION_AUDIT</code> and the service exposes them; the dashboard shows the latest six, and a dedicated, searchable audit screen is future work
Self-service password change	<code>AuthService.changePassword</code> (verifies the current password, applies the 8–64 character policy, wipes both arrays) is implemented, and its password-policy check is unit-tested, but it has no screen. Today a School Admin changes a password with Reset password on the Users & Roles screen; a small “Change my password” dialog in the sidebar would complete it
Arabic rendering	JavaFX places the space between an Arabic word and a following Western number on the wrong side. Dates, times and fractions are mitigated, and dashboard texts are phrased with the number first
Lockout scope	The failed-attempt counter is in memory per running application; a database-backed counter would share the lock across lab PCs
RBAC boundary	Enforced in the Java services with one schema account (§2.7.1); per-user database identities or a middle tier would close it
Distribution	Portable Windows x64 edition with an embedded Temurin 21 JRE is provided; the cross-platform bundle covers Linux x64 and the build machine’s macOS (Intel Macs need a build on an Intel Mac); an MSI installer (<code>j package</code>) with Start-menu shortcuts is a possible next step
Timetabling	Exam dates are entered by the control office; automatic timetable generation (graph colouring by shared students) is a natural extension

9. Appendix

9.1 Seed data

Entity	Content
Roles / users	4 roles; <code>admin</code> , <code>control</code> , <code>head</code> , <code>teacher</code>
Departments	8 (Mathematics, Physics, Chemistry, Biology, Arabic, English, Computer Science, Social Studies)
Courses	12 across Grades 10–12
Rooms	10; A104 under maintenance; HALL-1 with 60 exam seats
Exam periods	First 09:00–11:00, Second 12:00–14:30
Teachers	24 (3 per department) with daily/weekly caps
Students	210 — 70 per grade, sections A/B, every 15th with special needs
Exams	9 between 10 and 13 January 2027, including two exams sharing one slot
Relations	5 teacher–student relations (degrees 1–4)

9.2 Repository layout

```

|— pom.xml                build: fat jar; profile dist = lab bundle
|— run-app.bat / .sh      launchers (runtime\, JAVA_HOME, PATH)
|— packaging/            build-windows-portable.sh, build-docs-pdf.sh
|— README.md, DOCUMENTATION.md
|— config/               application.properties.example
|                        (the real file is git-ignored)
|— database/             00_drop_all ... 05_smoke_test,
|                        install.sql, install-docker.sh / .bat
|— src/main/java          121 source files (see §2.3)
|— src/main/resources     FXML, CSS, i18n bundles, logback.xml,
|                        application.properties
|— src/test/java          12 test classes, 84 test cases
|— src/assembly/         dist.xml: lab bundle layout

```

9.3 Glossary

Term	Meaning
Exam slot	an exam's date plus its period's start/end time
Exam capacity	seats usable during exams (spacing), $\leq$ regular capacity
Head of committee (رئيس لجنة)	person in charge of one exam room
Proctor (ملاحظ)	supervising teacher
Hours balance	accumulated supervision hours used for fair distribution
Soft delete	archiving by flag; the row stays for history and reports